BUSINESS ANALYTICS

Complete Learning Guide

From business metrics to evidence-based decisions

Designed for self-paced learning. Each module includes explanations, worked calculations, workplace scenarios, common mistakes and knowledge checks—not just definitions.

Assessment guidance: 70% required to pass; retakes are available. Learn the reasoning behind each answer rather than memorising terms.

How to use this guide

Read one module at a time. For each idea, ask what decision it supports, which evidence is required, and what could make the conclusion misleading. Work through the calculations yourself before checking the explanation.

Learning habit	How to apply it
Start with a decision	Clarify who will act and what they need to decide.
Define metrics	Specify formula, population, period and source.
Make assumptions visible	Separate observed facts from estimates and hypotheses.
Compare fairly	Use consistent time windows, definitions and populations.
Communicate action	State conclusion, evidence, risk, owner and next step.

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Module 1: How Businesses Are Measured

Revenue, cost, margin and the metrics leadership watches.

What you should be able to do

• Read a simple profit-and-loss structure.

• Distinguish leading and lagging indicators.

• Choose a north-star metric aligned with customer value.

1. Revenue, costs and profit

Revenue is the income earned from selling goods or services before costs are deducted. Cost of goods sold (COGS) captures direct costs associated with delivering what was sold. Gross profit = revenue – COGS. Operating expenses include items such as sales, marketing, administration and research. Operating profit deducts operating expenses from gross profit; net profit also reflects items such as interest and taxes.

2. Gross margin

Gross margin percentage = (revenue – COGS) ÷ revenue × 100. If revenue is ■10,00,000 and direct costs are ■6,00,000, gross profit is ■4,00,000 and gross margin is 40%. Margin is a percentage; gross profit is a currency amount. Higher margin is not automatically better if volume, customer value or strategic goals deteriorate.

3. Leading and lagging indicators

A leading indicator may change before a business outcome and provide an early signal, such as qualified sales pipeline or product activation. A lagging indicator confirms a result after it occurs, such as quarterly revenue or realised churn. Leading indicators are useful but imperfect predictors; validate whether they actually relate to the outcome.

4. North-star metric

A north-star metric captures the core value a business delivers to customers and helps align teams. For a collaboration product, it might be teams completing meaningful collaborative work each week; for a marketplace, successful transactions may be more meaningful than app downloads. A good metric is understandable, influenceable, tied to value and supported by guardrails.

5. Read a simple P&L

Example: revenue ■10 lakh – direct costs ■6 lakh = gross profit ■4 lakh. If operating expenses are ■2.5 lakh, operating profit is ■1.5 lakh before interest and tax. Always check whether figures cover the same period and whether costs are classified consistently.

Worked example / workplace scenario

Scenario: A subscription company sees signups rise but paid renewals fall. Signups may be a leading indicator, while realised renewal revenue and churn are lagging outcomes. Track activation and early usage to diagnose the funnel, but do not assume more signups automatically mean a healthier business.

Common mistakes to avoid

• Confusing revenue with profit or cash flow.

• Comparing margin percentages without checking cost definitions.

• Choosing a north-star metric that is easy to inflate but not tied to customer value.

Check your understanding

Q1. Revenue is ■8 lakh and direct costs are ■5 lakh. What is gross margin?

Q2. Give one possible leading and one lagging indicator for a subscription business.

Q3. What makes a north-star metric useful?

Suggested answers

A1. Gross profit is ■3 lakh; gross margin is ■3 lakh ÷ ■8 lakh × 100 = 37.5%.

A2. Leading: activation or qualified pipeline. Lagging: realised renewal rate or revenue.

A3. It reflects core customer value and helps align teams around a meaningful outcome.

Remember: A useful business analysis links a clear decision to consistent metrics, defensible assumptions and practical action.

Module 2: Requirement Gathering

Finding out what stakeholders actually need.

What you should be able to do

• Run a structured stakeholder interview.

• Separate stated wants from underlying needs.

• Write requirements that can be verified.

1. Identify stakeholders

Stakeholders include end users, managers, customers, operations teams, compliance, IT and anyone affected by the outcome. Different stakeholders may define success differently. Map their influence, needs, concerns and decision authority.

2. Discover the underlying need

A request for “a dashboard” is a proposed solution, not necessarily the underlying need. Ask what decision is difficult today, who makes it, how often, what evidence is missing, and what action will follow. Ask for recent examples and current workarounds.

3. Functional requirements

A functional requirement describes what the system must do. Example: “The report must allow a manager to filter completed orders by date range and region.” Make the behaviour observable and testable.

4. Non-functional requirements

These describe quality attributes or constraints: performance, security, availability, accessibility, reliability, scalability or auditability. Replace vague words such as “fast” with measurable thresholds, such as “95% of report requests complete within two seconds under the agreed load.”

5. Acceptance criteria

Acceptance criteria define the conditions under which a requirement is accepted. They should be specific, testable and agreed before delivery. Example: given a valid date range, when the manager selects a region, then the report displays only completed orders in that region and its total reconciles with the source.

6. Prioritisation and scope

Separate must-have needs from optional enhancements. Record assumptions, dependencies, out-of-scope items and unresolved questions. Confirm the final interpretation with stakeholders before implementation.

Worked example / workplace scenario

Example interview prompts: “What are you trying to decide?” “How do you do this today?” “What happens when it goes wrong?” “Which users need access?” “What would a successful result look like?” “How will we verify it?”

Common mistakes to avoid

• Accepting the first requested solution without exploring the problem.

• Writing requirements with words like easy, fast or secure but no criteria.

• Leaving stakeholders to discover scope differences at acceptance time.

Check your understanding

Q1. A user asks for a dashboard. What should you ask first?

Q2. Is response time a functional or non-functional requirement?

Q3. What do acceptance criteria provide?

Suggested answers

A1. Ask which decision or task is difficult and what outcome the dashboard should enable.

A2. Non-functional, because it describes performance quality.

A3. Specific conditions that can be checked to determine whether the requirement is met.

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Module 3: Process Mapping & Gap Analysis

Seeing the process as it operates before redesigning it.

What you should be able to do

• Map a current-state process end to end.

• Identify bottlenecks, rework loops and handoff losses.

• Compare current performance with a realistic future state.

1. Map the current state

Document how work actually happens—not only how a policy says it should happen. Include triggers, roles, systems, decisions, exceptions, queues, wait times and rework. Validate the map with people who perform the work.

2. Find bottlenecks

A bottleneck is the step whose capacity constrains overall throughput. A team may complete 100 cases daily, but if final approval handles only 60, work accumulates before approval. Adding capacity elsewhere may not improve end-to-end throughput.

3. Handoffs and rework

A handoff transfers work between people, teams or systems. It can create delays when ownership is unclear or information is incomplete. Rework occurs when a step must be repeated due to errors, missing details or failed checks. Measure frequency and time, not only anecdotes.

4. Gap analysis

Compare current performance with a defined target. Example: current average approval time is 4 days; target is 2 days; gap is 2 days. Break the gap into likely drivers such as queue time, missing documents and approval capacity.

5. Design a future state

Propose changes based on evidence, such as validating forms earlier, clarifying ownership, automating routine checks or changing queue rules. Estimate benefits, costs, risks and dependencies. Define measures that will show whether the change worked.

6. Validate improvement

Use baseline metrics and monitor after implementation. Consider both speed and quality: reducing handling time is not success if error rates or customer complaints rise.

Worked example / workplace scenario

Scenario: Applications take six days to approve. A process map shows one day of work and five days waiting for a manager who reviews requests only twice a week. The likely constraint is approval scheduling, not data entry. Test a daily review queue and measure cycle time, backlog and error rate.

Common mistakes to avoid

• Mapping the ideal process instead of the real one.

• Assuming the slowest-looking task is always the system bottleneck.

• Optimising speed without tracking errors, quality or customer impact.

Check your understanding

Q1. What is a bottleneck?

Q2. How is a handoff different from rework?

Q3. Why define a baseline before changing a process?

Suggested answers

A1. The capacity constraint limiting end-to-end throughput.

A2. A handoff transfers work; rework repeats a task to correct or redo it.

A3. To compare outcomes fairly and determine whether the change improved performance

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Module 4: Unit Economics

Whether the business makes money on each customer or transaction.

What you should be able to do

• Calculate contribution per unit or customer.

• Relate customer acquisition cost to lifetime value.

• Explain contribution margin and break-even volume.

1. Variable and fixed costs

Variable costs change with each unit or customer, such as packaging or payment fees. Fixed costs remain relatively stable over a relevant range and period, such as office rent. Some costs are mixed or step-fixed, so classify them carefully for the decision.

2. Contribution margin

Contribution per unit = selling price – variable cost per unit. Contribution margin percentage = contribution per unit ÷ selling price × 100. Contribution first covers fixed costs; after those are covered, additional contribution contributes to profit.

3. Customer acquisition cost (CAC)

CAC = relevant sales and marketing acquisition spend ÷ number of new customers acquired in the same period or cohort. Define which costs and customers are included. Avoid comparing a broad spend period with a different acquisition window without explaining the mismatch.

4. Lifetime value (LTV)

LTV estimates the contribution a customer generates over the relationship, not merely total revenue. A simple model might use average contribution per period multiplied by expected active periods, adjusted for churn, discounts or service costs. LTV is an estimate, so assumptions and cohort differences matter.

5. LTV:CAC and payback

Compare LTV with CAC to assess whether acquisition economics may be sustainable. A higher ratio is not enough on its own: payback timing, cash needs, churn, margin and uncertainty matter. Payback period is how long cumulative contribution takes to recover acquisition cost.

6. Break-even point

Break-even units = fixed costs ÷ contribution per unit. If fixed costs are ■2,00,000 and contribution per unit is ■500, break-even volume is 400 units. This assumes the price, variable cost and fixed-cost structure remain reasonably stable over the relevant range.

Worked example / workplace scenario

Worked example: A product sells for ■1,200 and has variable costs of ■700. Contribution per unit is ■500; contribution margin is 41.7%. With fixed costs of ■2,00,000, break-even volume is 400 units. If acquisition spend is ■1,00,000 and 200 new customers are acquired, CAC is ■500 per customer—assuming the spend and acquisition period are aligned.

Common mistakes to avoid

• Using revenue instead of contribution for LTV.

• Dividing acquisition spend by total customers rather than new customers when calculating CAC.

• Treating break-even estimates as exact when costs or prices change with volume.

Check your understanding

Q1. Price ■800, variable cost ■500: contribution per unit?

Q2. Spend ■60,000 acquires 120 customers: CAC?

Q3. Fixed costs ■90,000 and contribution ■300: break-even units?

Suggested answers

A1. ■300.

A2. ■500 per customer.

A3. 300 units

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Module 5: Forecasting & Scenario Planning

Modelling what might happen, with honest ranges.

What you should be able to do

• Build a forecast from operational drivers.

• Create base, upside and downside scenarios.

• Test which assumptions most affect the outcome.

1. Baseline

A baseline is the reference projection used to compare alternatives and actual results. It should state the period, data used, assumptions and whether it represents current performance or expected performance without a proposed change.

2. Driver-based forecasts

Build results from underlying drivers instead of applying one unexplained growth rate. For example: revenue = active customers × purchase frequency × average order value. For a subscription business, revenue may depend on opening customers, new customers, churn, plan mix and price.

3. Scenarios

Base, upside and downside scenarios use coherent sets of assumptions. An upside case might combine stronger conversion with moderate acquisition costs; a downside case might combine lower conversion and higher churn. Do not combine every optimistic assumption unless there is a defensible reason.

4. Sensitivity analysis

Change one assumption while holding others constant to understand its influence. If profit changes dramatically with churn but barely changes with website hosting cost, churn deserves more attention. Sensitivity analysis does not tell you which value will occur; it shows dependency.

5. Ranges and uncertainty

Use ranges where forecasts are uncertain. Separate facts, assumptions and management choices. State what evidence would cause the forecast to be updated, and compare actual results against forecast drivers rather than only the final total.

6. Forecast governance

Assign an owner to key assumptions, record when they were updated, and review forecast accuracy. Avoid false precision: a forecast of ■10,43,281 may imply more certainty than the underlying assumptions support.

Worked example / workplace scenario

Worked example: A business has 2,000 active customers, each placing 1.5 orders per month at an average value of ■800. Estimated monthly revenue = 2,000 × 1.5 × ■800 = ■24,00,000. If order frequency falls to 1.3 while other drivers stay constant, revenue becomes ■20,80,000. That assumption alone changes revenue by ■3,20,000.

Common mistakes to avoid

• Forecasting from a single growth percentage without explaining drivers.

• Changing many assumptions at once during sensitivity analysis.

• Reporting one precise number without explaining uncertainty.

Check your understanding

Q1. What is a baseline?

Q2. What does sensitivity analysis test?

Q3. Why should scenarios use coherent assumptions?

Suggested answers

A1. The reference projection used for comparison.

A2. How much an output changes when a particular assumption changes.

A3. Unrelated combinations of best/worst assumptions may create implausible outcomes.

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Module 6: Customer & Market Analysis

Understanding demand before designing a response.

What you should be able to do

• Segment a market by need and behaviour.

• Compare competitors systematically.

• Estimate a market opportunity using a defensible method.

1. Market segmentation

Divide customers into groups that differ in needs, behaviour, context or willingness to pay. Useful dimensions include use case, frequency, company size, geography, lifecycle stage and channel. A segment should be measurable, reachable, sufficiently valuable and meaningfully different.

2. Competitor analysis

Compare alternatives on criteria customers care about: price, features, reliability, service, ease of use, distribution and trust. Include direct competitors and substitutes. Use comparable evidence and distinguish public facts from assumptions.

3. Total addressable market (TAM)

TAM estimates the full demand for a relevant product or service if it could serve the entire target market. A top-down method starts with a market estimate and narrows it; a bottom-up method estimates reachable customers × realistic annual spend. Explain scope, geography, time period and assumptions.

4. SAM and SOM

Serviceable available market (SAM) narrows TAM to the portion the product and business model can serve. Serviceable obtainable market (SOM) estimates the share realistically capturable given competition, capacity and time. These are estimates, not guaranteed revenue.

5. Value proposition

A value proposition states who the product serves, what need it addresses, what benefit it delivers and why the customer should choose it over alternatives. It should be supported by customer evidence rather than internal preference.

6. Churn rate

A simple customer churn rate is customers lost during a period divided by customers at the start of that period, subject to a clearly stated definition. Exclude or separately report new customers when appropriate. Revenue churn and customer churn answer different questions.

Worked example / workplace scenario

Worked example: If a realistic target segment contains 50,000 eligible businesses and annual spend per business is ■24,000, a simple bottom-up TAM estimate is 50,000 × ■24,000 = ■12 crore per year. A realistic SAM/SOM will be smaller and needs explicit constraints and evidence.

Common mistakes to avoid

• Using all people in a country as a market without defining eligible buyers.

• Treating a competitor feature list as proof of customer preference.

• Mixing customer churn and revenue churn.

Check your understanding

Q1. What is one useful basis for segmentation?

Q2. How does SOM differ from TAM?

Q3. Start with 1,000 customers and lose 50 during the month: simple churn rate?

Suggested answers

A1. Need, behaviour, use case, geography or willingness to pay.

A2. SOM is the realistically obtainable portion; TAM is the full relevant opportunity.

A3. 5%, assuming the definition and period are consistent.

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Module 7: Dashboards for Decision Makers

Reporting that shortens the path to a decision.

What you should be able to do

• Design a management dashboard around decisions.

• Frame actuals against targets and comparisons.

• Set a useful reporting cadence and drill-down path.

1. Start with the decision

A dashboard should help a specific audience make a recurring decision. Identify the audience, decision, metrics, update frequency and action thresholds before choosing charts.

2. Actual, target and variance

Variance = actual – target. If actual revenue is ■9 lakh and target is ■10 lakh, variance is –■1 lakh. Variance percentage = (actual – target) ÷ target × 100 = –10%. For costs, a positive variance may be unfavourable; label favourable/unfavourable based on business meaning, not just the sign.

3. Drill-down

A drill-down moves from a summary to detail that may explain it: total revenue → region → product category → individual orders. Keep filters and definitions consistent so users can understand how detail reconciles to the headline.

4. Reporting cadence

Set the reporting rhythm to match how quickly the metric changes and how often a decision can be made. Operational queues may need daily review; strategic financial performance may be monthly or quarterly. Frequent reporting is not useful if data is unstable or no action can follow.

5. Dashboard layout

Put the most important KPIs first, followed by trends and breakdowns. Include period, target, previous-period comparison, data freshness and metric definitions. Avoid overcrowding and ensure a user can identify the most important exception quickly.

6. Alerts and ownership

Define thresholds, who receives alerts, and what action is expected. A dashboard without an owner or response process may report problems without resolving them.

Worked example / workplace scenario

Worked example: Actual delivery time is 3.4 days against a target of 3.0 days. Variance is +0.4 days; because lower time is better, this is unfavourable. A drill-down by warehouse and carrier can help identify where delays concentrate.

Common mistakes to avoid

• Showing a number without target, period or definition.

• Calling all positive variances good, including costs and delays.

• Building a dashboard with no clear audience or action.

Check your understanding

Q1. Actual revenue ■90,000; target ■1,00,000. What is variance percentage?

Q2. What is drill-down?

Q3. How should reporting cadence be chosen?

Suggested answers

A1. (90,000 – 100,000) ÷ 100,000 × 100 = –10%.

A2. Moving from summary values into more detailed records or segments.

A3. Based on metric volatility and the frequency of decisions it supports.

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Module 8: Root Cause Analysis

Getting past the symptom to a testable explanation.

What you should be able to do

• Use five whys and issue trees appropriately.

• Structure causes into useful, non-overlapping categories.

• Test hypotheses and avoid confusing correlation with causation.

1. Define the problem precisely

Describe the symptom with metric, population, period and comparison. “Customer service is bad” is vague; “the share of tickets resolved within 24 hours fell from 82% to 65% in September” is actionable.

2. Issue trees

An issue tree breaks a question into subquestions. For a revenue decline, branches might be number of customers, purchase frequency and average order value. Good branches are mutually exclusive and collectively cover the main possibilities, reducing overlap and gaps.

3. Five whys

Repeatedly ask why an observed effect occurred, using evidence at each step. The technique helps explore a causal chain but does not guarantee the final answer is the root cause. Stop when a cause is specific, actionable and supported, and check for multiple contributing causes.

4. Hypotheses

A hypothesis is a specific, testable proposed explanation stated before analysis. Example: “Checkout conversion declined because payment failures increased on mobile.” Define what data would support or contradict it and what alternative explanations must be checked.

5. Correlation trap

Two variables may move together because of a third factor, reverse causation or coincidence. To support causal claims, consider timing, comparison groups, experiments or other suitable causal methods. Do not state that an intervention caused improvement merely because the measures moved together.

6. Prioritise investigation

Rank possible causes by evidence, expected impact, likelihood and cost of testing. Start with the explanations that can be checked efficiently and that would change the decision.

Worked example / workplace scenario

Scenario: Revenue drops. An issue tree separates customer count, order frequency and average order value. Data shows customer count is stable but order frequency fell in one channel. A hypothesis about a recent checkout change can then be tested using error logs, device segments and a suitable before/after or controlled comparison.

Common mistakes to avoid

• Asking “why” repeatedly without checking evidence.

• Creating overlapping issue-tree branches that double-count causes.

• Treating a plausible hypothesis as a confirmed root cause.

Check your understanding

Q1. What makes a hypothesis testable?

Q2. Why use an issue tree?

Q3. Does correlation prove causation?

Suggested answers

A1. It states a specific explanation and identifies observations that could support or contradict it.

A2. It decomposes a problem into structured branches and helps find gaps or overlaps.

A3. No; alternative explanations and study design must be considered.

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Module 9: Building a Business Case

Turning analysis into an approved decision.

What you should be able to do

• Structure a case from problem to decision request.

• Quantify benefits, costs, ROI and payback.

• Present risks with credible mitigations.

1. Business-case structure

A clear case states the problem, evidence, options considered, recommendation, expected benefits, costs, assumptions, risks, implementation plan and decision requested. Make the requested approval explicit.

2. Quantify benefits and costs

Estimate one-time implementation costs and ongoing costs. Benefits may include incremental revenue, avoided cost, capacity released or reduced risk. Avoid counting the same benefit twice and distinguish cash savings from time saved unless the latter truly reduces cost or enables additional value.

3. Return on investment (ROI)

A common simple formula is ROI = (net benefit ÷ investment cost) × 100, where net benefit = total benefit – total cost over a defined period. Organisations use different conventions, so state the formula, included costs and time horizon.

4. Payback period

Payback is the time until cumulative benefits equal initial investment. With a ■6 lakh initial cost and a steady net benefit of ■1 lakh per month, simple payback is six months. If benefits vary by month, calculate cumulative cash flow instead of dividing by a single average.

5. Risks and mitigation

A risk is an uncertain event that may affect outcomes. Assess likelihood and impact, name an owner, and define mitigation or contingency actions. Mitigation reduces likelihood or impact; it does not guarantee that the risk disappears.

6. Compare alternatives

Include a credible do-nothing baseline and alternatives. A lower-cost option may have lower benefit or higher operational risk. Make trade-offs explicit and show which assumptions drive the recommendation.

Worked example / workplace scenario

Worked example: A project costs ■4 lakh and is expected to generate ■6 lakh in benefits over a defined period, with no other costs included. Net benefit = ■2 lakh; simple ROI = ■2 lakh ÷ ■4 lakh × 100 = 50%. State the time horizon and whether ongoing costs are included.

Common mistakes to avoid

• Reporting gross benefits as if they were net benefits.

• Using ROI without a defined time horizon.

• Ignoring implementation, maintenance, training or change-management costs.

Check your understanding

Q1. Investment ■2 lakh; net benefit ■50,000. Simple ROI?

Q2. What is payback period?

Q3. What should a mitigation plan specify?

Suggested answers

A1. 25%.

A2. The time until cumulative benefits recover the investment cost.

A3. An action, owner, timing and how it reduces likelihood or impact.

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Module 10: Communicating Recommendations

Being understood by people who did not do the analysis.

What you should be able to do

• Lead with the answer and support it with evidence.

• Adapt detail to an executive or technical audience.

• Record decisions and respond to challenges with evidence.

1. Answer-first structure

Lead with the recommendation or conclusion, then explain the most important evidence, implications, risks and next steps. Do not force decision-makers to search through methodology before they know the point.

2. Adapt to the audience

Executives often need the decision, impact, cost, risk and timing. Implementers need operational details, dependencies and acceptance criteria. Technical reviewers may need methods, assumptions, data definitions and validation. The facts stay the same; the depth and framing change.

3. Separate facts from interpretation

Label observed results, assumptions, hypotheses and recommendations clearly. State the level of confidence and what new evidence could change the recommendation.

4. Use appendices well

Keep detailed calculations, definitions, query logic and supporting charts in an appendix when they would interrupt the main narrative. The appendix should allow reviewers to verify the analysis, not hide material caveats.

5. Decision log

Record what was decided, by whom, when, based on which evidence, and what actions follow. Include unresolved risks and review dates. A decision log supports accountability and prevents teams from reopening settled questions without new information.

6. Handle challenge constructively

When questioned, return to definitions, data and assumptions. Acknowledge valid limitations, correct mistakes, and distinguish what is known from what is uncertain. Do not abandon a well-supported conclusion merely because it is challenged, but remain willing to update it when evidence changes.

Worked example / workplace scenario

Example recommendation: “Approve a four-week pilot of earlier payment-failure alerts for mobile checkout. Mobile payment failures rose from 2% to 5% while traffic was stable. The pilot will track successful checkout rate and false alerts. The evidence is consistent with a payment issue but does not yet prove the alert will recover all lost orders.”

Common mistakes to avoid

• Burying the recommendation under background detail.

• Using confidence language that is stronger than the evidence.

• Failing to record the decision owner and next action.

Check your understanding

Q1. What should come first in an answer-first presentation?

Q2. What belongs in an appendix?

Q3. What should a decision log capture?

Suggested answers

A1. The conclusion or recommendation.

A2. Supporting detail needed for verification but not essential to the main narrative.

A3. Decision, owner, date, evidence, follow-up actions and unresolved risks.

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Final Assessment-Style Practice Quiz

Choose the single best answer for each question. This practice quiz is not the platform's official assessment. A score of 14/20 equals 70%.

1. Revenue is ■10 lakh and direct costs are ■6 lakh. Gross margin is:

A. 40%

B. 60%

C. 4%

D. 166%

2. Which is most likely a lagging indicator?

A. New user activation

B. Qualified pipeline

C. Revenue already earned last quarter

D. Product demo requests

3. A north-star metric should primarily reflect:

A. A number that is easy to inflate

B. Core customer value delivered

C. Only marketing activity

D. The largest possible number

4. “95% of reports load within two seconds” is primarily:

A. A functional requirement

B. A non-functional requirement

C. A stakeholder

D. A business case

5. Acceptance criteria are used to:

A. Decorate a report

B. Verify whether a requirement is satisfied

C. Estimate market size

D. Calculate gross margin

6. A process step limits the throughput of the entire process. It is a:

A. Handoff

B. Bottleneck

C. North-star metric

D. Baseline

7. Actual approval time is 4 days; target is 2 days. The gap is:

A. 2 days

B. 6 days

C. 50 days

D. 0 days

8. Price ■1,200 and variable cost ■700. Contribution per unit is:

A. ■500

B. ■700

C. ■1,200

D. ■1,900

9. Spend ■1,00,000 acquires 200 customers. CAC is:

A. ■200

B. ■500

C. ■2,000

D. ■20,000

10. Fixed costs ■2,00,000; contribution ■500 per unit. Break-even volume is:

A. 100

B. 200

C. 400

D. 1,000

11. Sensitivity analysis typically changes:

A. One assumption while holding others fixed

B. Every assumption at once

C. Only the chart colour

D. No assumptions

12. A defensible bottom-up TAM estimate often uses:

A. Eligible customers × realistic spend

B. Company headcount × salary

C. Only competitor slogans

D. Current profit × 100

13. Start with 1,000 customers and lose 50 during the period. Simple churn is:

A. 0.5%

B. 5%

C. 50%

D. 20%

14. Actual cost is ■1.2 lakh and target cost is ■1 lakh. The variance is:

A. ■20,000 favourable by default

B. ■20,000 unfavourable if lower cost was the target

C. ■2.2 lakh

D. 0%

15. A useful issue tree should:

A. Have overlapping branches

B. Break a problem into structured, non-overlapping parts that cover the question

C. Prove causation by itself

D. Replace data

16. Two metrics move together. What can you conclude?

A. One caused the other for sure

B. There is an association worth investigating

C. There is no relationship

D. An experiment is unnecessary

17. Investment cost ■4 lakh; net benefit ■2 lakh. Simple ROI is:

A. 25%

B. 50%

C. 100%

D. 200%

18. A steady net benefit of ■1 lakh/month recovers ■6 lakh after:

A. 3 months

B. 6 months

C. 12 months

D. 60 months

19. What should usually come first in an executive recommendation?

A. All raw data

B. The answer or recommendation

C. A long methods section

D. An appendix

20. A decision log should record:

A. Only dashboard colours

B. Decision, owner, evidence and follow-up

C. Every email password

D. Only the final metric

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Answer Key and Explanations

Q	Answer	Explanation
1	A	Gross profit is ■4 lakh; ■4 lakh ÷ ■10 lakh = 40%.
2	C	It confirms an outcome already realised.
3	B	It should align teams around meaningful customer value.
4	B	It specifies a performance-quality target.
5	B	They define observable conditions for acceptance.
6	B	The bottleneck constrains overall output.
7	A	Current minus target is two days.
8	A	Contribution equals price minus variable cost.
9	B	■1,00,000 ÷ 200 = ■500.
10	C	■2,00,000 ÷ ■500 = 400 units.
11	A	It isolates the effect of one input on the result.
12	A	It builds an estimate from an explicit customer base and spend.
13	B	50 ÷ 1,000 × 100 = 5%.
14	B	Actual exceeds target; for cost, that is normally unfavourable.
15	B	Good issue trees reduce overlap and gaps.
16	B	Correlation alone does not establish causality.
17	B	■2 lakh ÷ ■4 lakh × 100 = 50%.
18	B	Simple payback is cost divided by monthly net benefit.
19	B	Answer-first communication shortens the path to a decision.
20	B	It preserves the reasoning and accountability for action.

Quick Revision Sheet

Concept	Remember
Gross margin	(Revenue – direct costs) ÷ revenue × 100.
Leading / lagging	Leading may signal future outcomes; lagging confirms results already realised.
Requirements	Functional = what it does; non-functional = quality constraint; acceptance criteria = testable conditions.
Process analysis	Map actual work; find bottlenecks, handoffs, rework and gaps to target.
Unit economics	Contribution = price – variable cost; CAC = acquisition spend ÷ new customers; break-even = fixed costs ÷ contribution.
Forecasting	Driver-based projections, coherent scenarios, sensitivity checks and explicit assumptions.
Market analysis	Segment by need/behaviour; estimate TAM with transparent scope; distinguish TAM, SAM and SOM.
Dashboards	Show actual, target, variance, trend, period and drill-down; cadence should support decisions.
Root cause	Issue trees structure possibilities; five whys explore chains; hypotheses need evidence.
Business case	Compare benefits, costs, ROI, payback, alternatives, risks and mitigation.
Recommendations	Lead with the answer, support it, state limitations and record decisions.

Before presenting a recommendation

- Is the business question and decision clear?
- Are metric formulas, populations and periods consistent?
- Have I separated facts, assumptions and hypotheses?
- Are costs, benefits and uncertainty represented honestly?
- Does the recommendation identify an action and success measure?
- Are risks, owners and next steps explicit?

Final principle: Strong business analysis does not stop at a number. It explains what the number means, what may have caused it, what decision it supports, and what should happen next.